Project Proposal — Text-to-Image Generation with CLIP-Guided Latent Diffusion

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1. Problem & Goal

Generate images that faithfully match natural-language descriptions. We'll start from a pretrained latent diffusion model, measure how well its outputs match their prompts on everyday scenes, then improve that match by fine-tuning on a curated image–caption subset. **Success metric:** a higher CLIPScore and better text→image retrieval than the zero-shot baseline, plus a human preference study on held-out prompts.

2. Dataset — MS-COCO 2017 Captions (curated subset)

Source	COCO val2017: 5,000 images, ~25k human captions (5 per image)
Why COCO over Flickr30k	More object/scene variety (80 categories), and it's the standard benchmark for text-to-image FID/CLIPScore. The 1 GB val split fits our 1k–5k target without the 18 GB train download.
Cleaning	Normalize whitespace and punctuation; keep captions of 5–40 words that fit in 77 CLIP tokens and are ASCII; drop exact duplicates; drop images whose short side is under 256 px or whose aspect ratio is over 2:1; fix EXIF orientation and convert to RGB
Preprocessing	Resize the short side to 512 px and center-crop to 512×512. Keep one primary caption per image and the rest for augmentation. 90/10 train/val split (seed 42). Output is HF-`imagefolder` `metadata.jsonl`
Size	**4,926 pairs** (4,434 train / 492 val). We dropped 247 duplicate captions and 74 images that failed the size or shape checks

3. Model & Method

- **Text encoder:** CLIP ViT-L/14. Each prompt becomes a 77×768 per-token embedding (fed to the UNet through cross-attention) and a pooled 768-d embedding (used for scoring and retrieval).
- **Generator:** Stable Diffusion v1.5 (latent diffusion: VAE plus a UNet conditioned on text), fp16, DPM-Solver++ with 30 steps and CFG 7.5. It fits our 8 GB RTX 5060 GPUs.
- **Phase 1 (this milestone, done):** zero-shot baseline. Text→image Recall@1/5/10 is 0.67/0.91/0.97, and CLIPScore on real pairs is 26.1. Across the 5 sample prompt→image runs, the mean CLIPScore is 28.9 at about 5 s per image. The samples show composition failures: a bus with the wrong number of decks and a missing object.
- **Phase 2:** LoRA fine-tuning of the UNet's cross-attention layers on the training split (rank 4–16, about 3M trainable parameters).
- **Phase 3:** ablations on guidance scale, step count, and caption choice (primary vs. augmented), then a comparison with a CLIP-guided or prompt-weighting variant.
- **Evaluation:** CLIPScore, FID on the 500-image val split (`torchmetrics`), text→image Recall@K, and a small blind human preference study.

4. Roles

Member	Role	Owns
Gokul Chandar Kumaran	**Modeling lead & integration**	Diffusers pipeline, LoRA fine-tuning, and GPU/experiment configs, plus the repo, environment, and milestone reports
Loalith Madala	**Data lead**	Download, cleaning, and filtering scripts; dataset stats and EDA; caption augmentation
Yagna Praneeth Reddy Panguru	**Evaluation lead**	CLIPScore, FID, and Recall@K harness; human preference study; result plots and demo

All three members share the write-up and the final presentation.

5. Milestone 1 Deliverables & Risks

Delivered (all in the repo): the dataset pipeline (`scripts/preprocess.py`), the environment check (`scripts/check_env.py`, PyTorch with CUDA 12.8, Diffusers, and CLIP), the baseline notebook (`notebooks/01_text_to_embedding_baseline.ipynb`), and 5 sample prompt→image runs (`outputs/samples/grid.png`).

Risks: limited VRAM (mitigated with fp16, attention slicing, and LoRA instead of full fine-tuning); COCO caption noise (mitigated by filtering and choosing the best caption per image); dataset licensing (COCO images are under Flickr CC licenses, so we use them for research only).